CHAPTER 4
PC MEMORY

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Chapter Objectives

Upon completion of this chapter, the student will be able to:

- Distinguish between different physical memory types.
- View & manage memory using Windows OS.
- Troubleshoot memory problems.
- Install RAM modules on the motherboard.

Sub-topics

4.1 Introduction
4.2 ROM types
4.3 RAM types
4.4 View & Manage memory
4.5 Troubleshooting memory problems
4.6 Upgrading memory (RAM)

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4.1 Introduction

PC memory can be classified by:

1. Location
   (a) Main memory (mainly RAM) (b) Secondary memory
2. Application
   (a) RAM (b) ROM (c) Hybrid (mixture of RAM & ROM)
3. Purpose
   (a) Physical memory (b) Available memory

RAM (Random Access Memory)

This is **volatile** memory which requires constant power (refreshing) to retain the content e.g. CMOS RAM.
If power is off, all information in RAM is lost i.e. switches change from on states (**1s**) to off states (**0s**).
It is used to hold temporary instructions and data for manipulation while the system is running.
Random means the CPU can access or place data to and from any RAM on the system.

ROM (Read Only Memory)

This is **non-volatile** memory used to store information permanently. ROM chips do not require constant power (refreshing) to retain the content. ROM consists of transistors that are manufactured permanently in the **on** or **off states**.
Once these transistors have been set, they cannot be changed e.g. ROM BIOS.

Physical memory

This refers to the capacity of RAM installed in a PC i.e. CPU reads and executes instructions in programs stored in physical memory.
Common values of physical memory:

1. Old systems 128, 256, 512 &1024 MB (e.g. 2 RAM modules @ 512 MB)
2. New systems 1 GB (1024 MB), 2 GB & 4 GB of physical memory

Available memory

-This refers to physical memory (RAM) available for use by the PC, after booting (loading OS). The PC uses this memory to load OS, applications and data files.

-Each time you open a program, it is loaded into physical memory which reduces it.
Large programs using graphics and sound (e.g. Adobe Photoshop) take up more physical memory than small programs which use only text (e.g. word processors).
-To open a large program, you should first quit other programs that are running on the PC. For example, use Task manager. How? Use Ctrl+Alt+Del.

How can you increase available memory?
1. Uninstall system files that are often installed by utility programs.
2. Increase physical memory by installing more RAM in memory slots.

Bus cycle
Refers to each communication between CPU and memory. CPU addresses amount of memory in a single bus cycle.

The memory controller (chip on motherboard):
- Handles the movement of data to and from the CPU and the system memory banks
- Responsible for the integrity of the data as it is swapped in and out of memory

The 2 main methods used are:
- Parity check
- ECC (Error Correction Coding)

Parity check
-This a simple method used to ensure data integrity (detect errors) by adding a **parity bit** (extra 9th bit) in addition to 8-bit bus cycle.
-There are 2 types of parity check:
1. Even parity 2. Odd parity

ECC (Error Correction Coding)
-This can detect more errors than the simple parity check method by adding extra information about bits.
See Data Communications & Networking for details.

Comparison of PC memory
1. Retention
   (a) Volatile (short term storage) (b) Non-volatile (long term storage)
2. Access time (Speed)
   (a) High speed (b) Slow speed
3. Capacity
   (a) Small capacity in MB (b) Big capacity in MB
4. Upgradability
   (a) Replaceable (b) Non-replaceable
5. Cost
   (a) Less expensive (b) More expensive

Access time
Also known as **speed**; is the response time the RAM takes to provide requested data to the memory controller and measured in nanoseconds (ns). For example, 60 ns is faster than 70 ns DRAM chip.

Summary

	Factor	Main memory	Secondary memory
1	Retention	Volatile Short term storage	Non-volatile Long term storage
2	Access time (Speed)	High speed Fast retrieval via parallel port	Low speed Slow retrieval via serial port
3	Capacity	Small capacity	Big capacity
4	Cost	More expensive	Less expensive
5	Upgradability	Replaceable	Replaceable

	Factor	RAM	ROM
1	Retention	Volatile Short term storage	Non-volatile Long term storage
2	Access time (Speed)	Fast retrieval via parallel port High speed	Slow retrieval via serial port Low speed
3	Capacity	Small capacity	Big capacity
4	Cost	Less expensive	More expensive
5	Upgradability	Replaceable	Non-replaceable

Remember:
- Do not mix RAM modules with different speeds in the same bank (set of several memory modules).
- Check the motherboard specifications for the recommended RAM chip speed.

4.2 ROM types

In the order of improvements, they are:

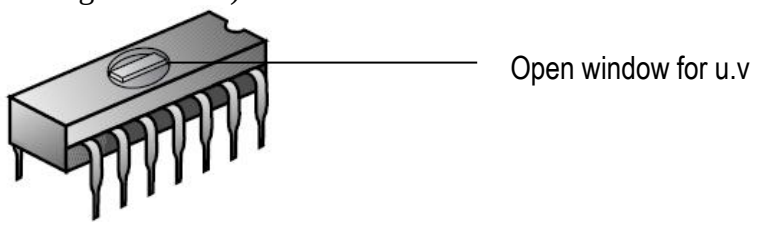
1. PROM 2. EPROM 3. EEPROM

PROM (Programmable ROM)

This is manufactured with all logical switches on (**1s**), and then the user can program it using a special device connected to PC through DB25 parallel port.
Programming involves changing some **1s** to **0s** using a high voltage electrical pulse, making the settings permanent. For example, **One Time Programmable (OTP) device** can be used to program PROM.
Disadvantage, It cannot be erased once it has been manufactured.

EPROM (Erasable PROM)

This can be set back to PROM using **U.V (ultra violet)** light outside a PC i.e. can be erased and programmed many times. It makes it more flexible, but more expensive than PROM.
A special frequency U.V light is shone through a small circular window on top of the chip (see diagram below).



The chip is easily identified by its small circular window and were used in old PCs e.g. IBM old PCs used EPROMs for their BIOS information.

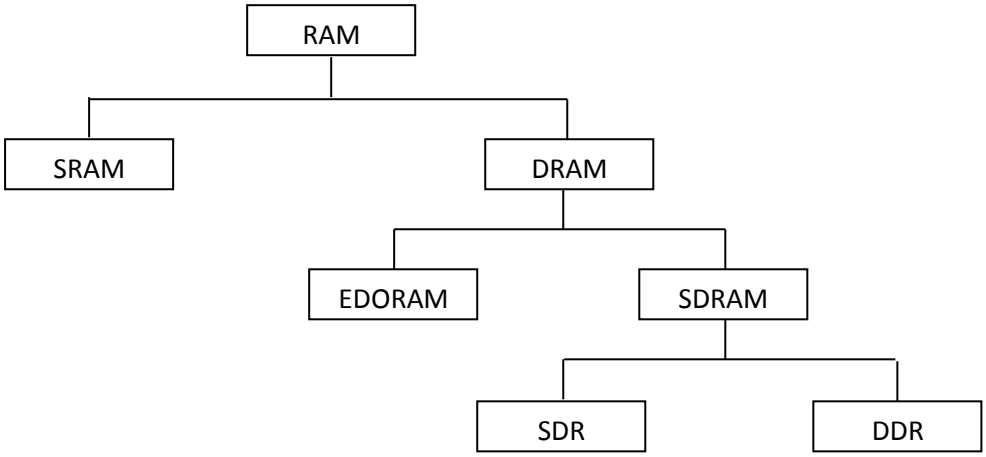
Disadvantage
You can inadvertently (without intention) erase information from the chip (becomes blank) by opening the PC case and exposing it to sun light (carries U.V).

Precaution!
Do not expose open window to sunlight (U.V) when opening system case (make sure open window is covered an opaque material).

EEPROM (Electrically EPROM)
Also known as **flash ROM** or **updatable ROM** for modern PCs e.g. **flash BIOS** used to keep information in PC. This can be erased and reprogrammed using a special sequence of **electric signals** to the chip while it is still in the circuit. It is used in most modern system ROM (e.g. industrial automation like robots) & in modern PCs as flash ROM.

Disadvantage, it's very expensive.

4.3 RAM types
In the stages of improvements as shown below:



The 2 main initial types of RAM were:
1. SRAM (Static RAM) 2. DRAM (Dynamic RAM)

Comparison
- Both are volatile because they lose their contents when the power is turned off.
- SDRAM is static because the information does not need a constant update (refreshing), while DRAM is dynamic because it needs constant update (refreshed thousands of times per second).
- SDRAM uses transistors, while DRAM uses microscopic capacitors (makes it cheaper) to store information.
- SRAM is faster because it does not need constant refreshing, but DRAM is more commonly used because it is cheaper than SRAM.

Summary

	Factor	SRAM	DRAM
1	Retention	No need for constant update (refreshing). Retains contents as long as power is on.	Needs constant update (refreshing) and so more power needed. Very short data life-time (approx. 4 milliseconds)
2	Access time (Speed)	Faster (10 ns)	Slower (60 ns)
3	Storage type	Transistors (binary registers).	Very small capacitors.
4	Capacity	Smaller capacity. (256 KB per chip)	Larger capacity.
5	Cost	More expensive.	Cheaper.
6	Application	Used in older PCs e.g. XT	Used in newer PCs.

The 2 main improved types of DRAM were:

1. EDORAM (Extended Data Output RAM)
2. SDRAM (Synchronous DRAM)

Comparison

- EDORAM is faster RAM which was used in Pentiums and became popular in 1995. Usually, it is a bit more expensive than regular DRAM.
- SDRAM is synchronized to the speed of memory bus and became popular in 1996 e.g. PC133 SDRAM runs at 133 MHz.
- SDRAM was developed to match the ever increasing processing speeds of the Pentium systems.

Advantage of SDRAM over EDORAM
The address bus does not have to wait for the memory bus working at different clock speed.

What happens if you install PC133 SDRAM in 100MHz bus?
The PC will function, but the PC133 SDRAM will only operate at the 100 MHz bus speed, thus reducing overall system performance.
The 2 types of SDRAM are:

1. SDR (Single Data Rate) 2. DDR (Dual Data Rate)

Advantage of DDR over SDR
It works at double the speed of SDR, which means increased throughput (data rate) and so increased overall PC speed.

Qn. What happens if you use a slower DDR2 module in a system of higher speeds?
Ans. The bus will run at the speed of the slowest memory in use.

Qn.
DDR with 100 MHz memory bus (bus clock rate) and data transfer rate of 8 mbps.
What is the maximum data rate of the module?
Soln.
DDR clock rate = memory bus clock rate x 2 = 100 MHz x 2 = 200 MHz (double)
Executes (reads & writes) at 200 MHz and transfer data rate to processor at
100 MHz

Maximum DDR data rate = DDR clock rate x data rate = 200 MHz x 8 = 1600 mbps.
-The 2 versions of DDR are:
1. DDR (older version) 2. DDR2 (newer version)
-DDR is the older version and DDR2 is an improvement over DDR
i.e. DDR2 operates at twice bus speed of DDR.
-DDR2 is used for high speed storage of working data in a PC or other digital electronic device.

Note:
- MDDR (Mobile DDR) is a type of memory used in some portable electronic devices, e.g. mobile phones.
- DDR operates at 2.5 V, but MDDR operates at 1.8 V (reduced power consumption).

DDR2 module with 240 pins

Disadvantage of DDR
More expensive because of higher speed and more difficult to assemble.

Comparison
- No. of pins i.e. DDR DIMM has 184 pins & DDR2 DIMM has 240 pins.
- No. of notches i.e. DDR DIMM has 1 notch & DDR2 DIMM has 2 notches.
- Pin density of DDR2 DIMM is slightly higher than DDR DIMM in desktops.
- DDR2 DIMMs do not have backward compatibility with DDR DIMMs.
- Faster DDR2 DIMMs are compatible with slower DDR2 DIMMs.
- However, faster DDR2 DIMM module runs at the slower module's speed.

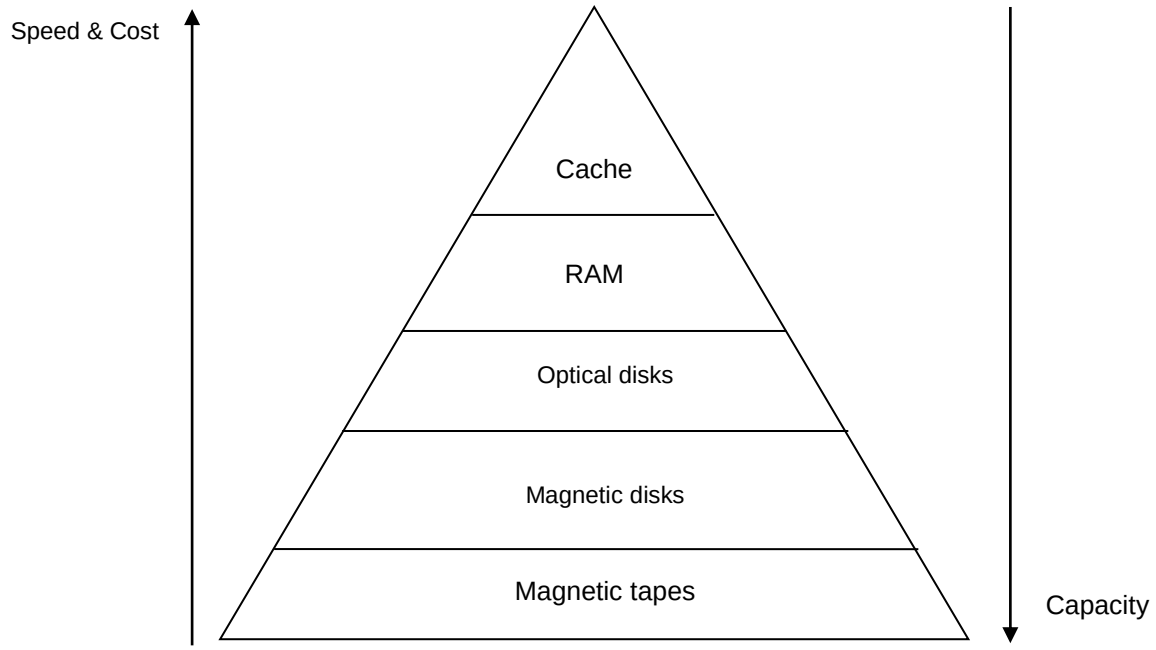
Specialised RAM types
1. VRAM 2. WRAM 3. Portable RAM 4.Cache memory

VRAM (Video RAM)
Special and more expensive type of RAM chip capable of:
- Performing reads and writes at the same time
- Being accessed by two devices simultaneously

It is commonly used as **video memory** i.e. by video card to store image data for processing.
Advantage
More VRAM means better quality of image displayed and higher resolution of image.

WRAM (Windows RAM)
Special type of RAM for Windows accelerator cards (AGP technology), developed by **Samsung** (South Korea) and similar to VRAM (but much faster). It has the ability to be read from or written to simultaneously and therefore also known as **dual-ported memory**.
Portable RAM
This type is used for portable PCs (laptops) and each manufacturer has different style and specifications.

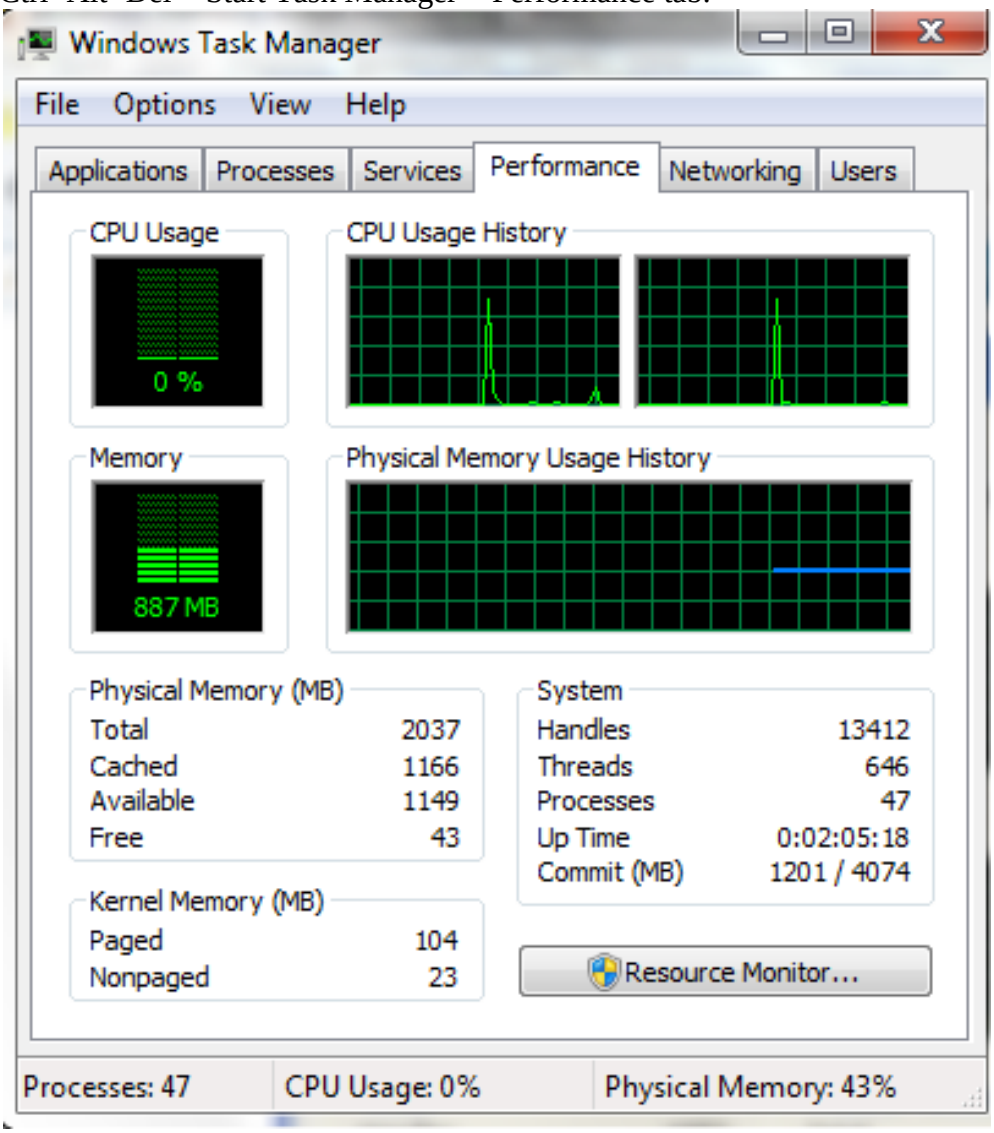
Cache memory
-This is a special memory which is much faster and more expensive than SRAM, but of small amount used as a buffer to reduce the time of memory access.
-There are 2 levels: 1. L1 (level 1) 2. L2 (level 2)
-Level 1 memory is within the microprocessor, while Level 2 memory is between microprocessor and main memory (RAM modules).
-Frequently accessed files (e.g. My Documents) are loaded in cache memory and accessed from here. This is known as **file caching** and speeds up access to a file.
See memory chip package types in Chap3 Motherboards.
Comparison of memory types



Note:
- Speed & Cost increase upwards, while Capacity increases downwards.
- Cache memory has the least capacity, but highest speed & most expensive per unit.
- Magnetic tapes have highest capacity, but lowest speed & cheapest per unit.

4.4 View & Manage memory
Windows 7 physical memory usage

Ctrl+Alt+Del > Start Task Manager > Performance tab.

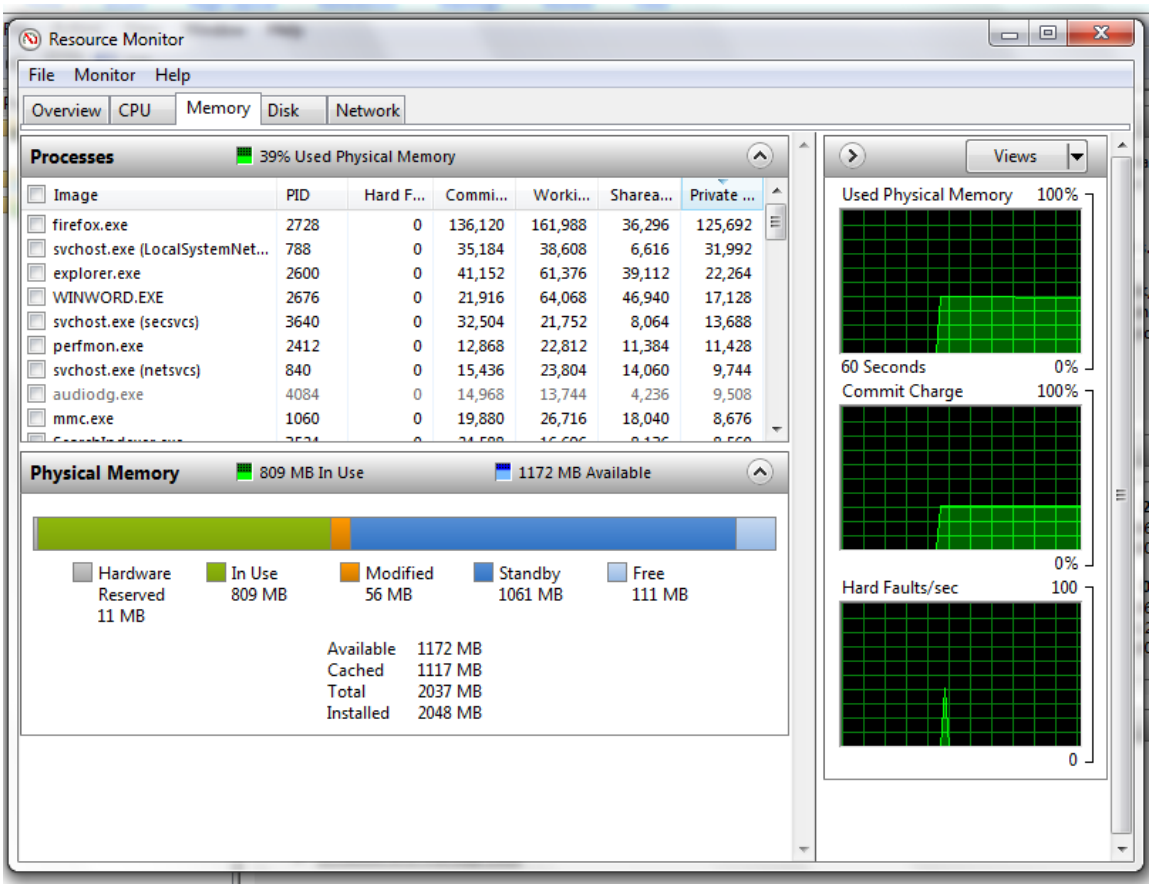


The Task Manager provides the following information:

- Total Physical Memory is the total RAM installed in the computer.
- Available Physical Memory is that part of installed RAM not allocated to program/process.
- Kernel Memory is the memory used by the operating system.
- Handles (object handles) is the number of instructions handled.
- Processes are programs running (currently active) in RAM.

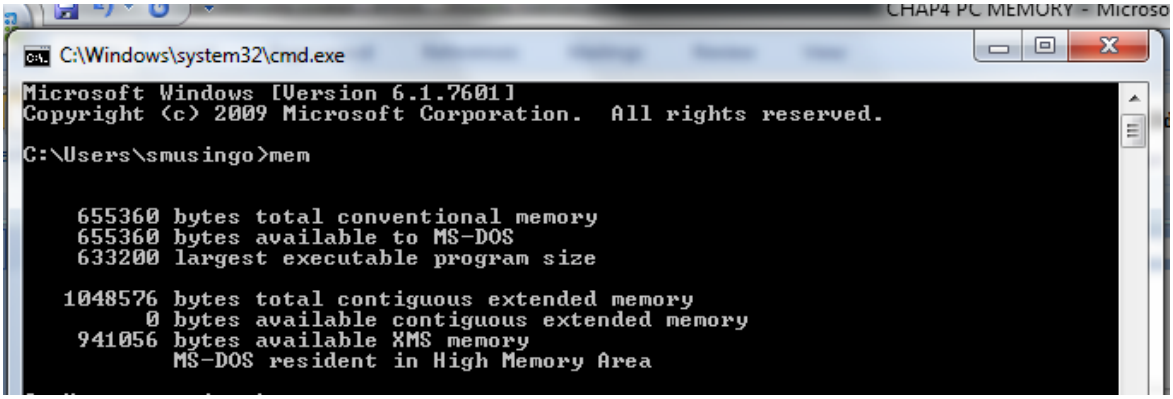
See Operating Systems Course Unit for details.

Click Start > Type “perform” in Search > Open Resource monitor > Memory.



MS DOS physical memory usage

At command prompt>: Type MEM & press Enter key. **mem.exe** program runs.



Note:

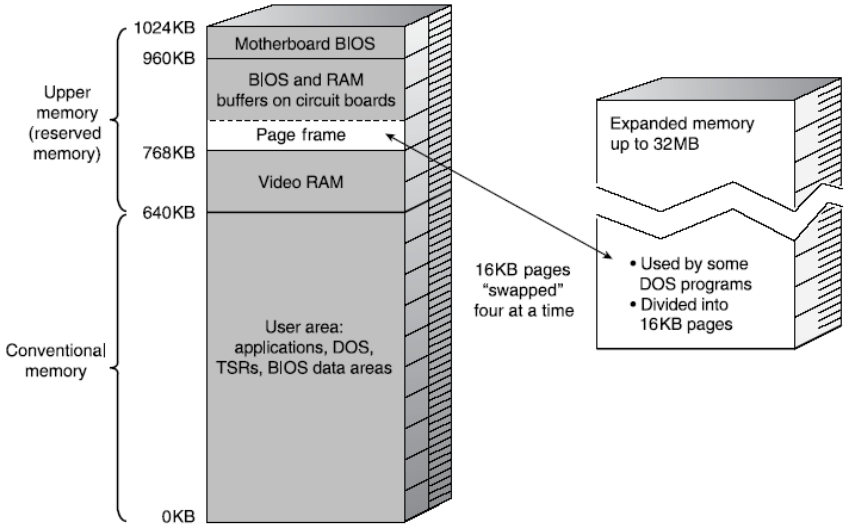
- MS DOS 6.22 was the last independently available MS DOS package marketed.
- All versions of Windows OS retain a shell (command line interface) with **cmd.exe** in Windows-NT family which replaced **command.exe**.

MS DOS memory management

Memory management includes:

1. To allocate memory to programs at their request.
2. To free memory for re-use when no longer needed.

Logical memory refers to the way physical memory (RAM) is represented by the operating system. MS DOS represents memory as stacks of 16 blocks each with a range of 5-digit **hexadecimal addresses** e.g. A0000 to AFFFF, B0000 to BFFFF and C0000 to CFFFF.



-Standard memory occupies the 1st 1MB (1024 KB) of main memory that DOS supports and consists of 2 types:

1. Conventional memory (0-640 KB) 2. Reserved memory (640-1024 KB)

-Extra memory is beyond standard memory and needed by some applications which require extra space. It consists of 2 types:

1. Expanded memory 2. Extended memory

Conventional memory

Also known as **basic memory**; consists of the lower 10 blocks of 640 KB (0 to 639 KB or 00000 to 9FFFF) that DOS supports in the memory map.
This memory is used for running programs, loading OS files and loading drivers.
OS communicates to hardware devices through their drivers e.g. sound card driver is loaded in conventional memory.
TSR (Terminate & Stay Resident) programs (e.g. anti-virus programs, disk caching programs) need to stay in conventional memory constantly.

Reserved memory

Also known as **upper memory** or **UMA (Upper Memory Area)**; consists of the upper 6 blocks of 384 KB (640 to 1024 KB or A0000 to FFFFF) that DOS supports in the memory map. It is fast & efficient memory reserved for use by some devices in the PC to store data for direct access by the microprocessor. Various sections of reserved memory are usually allocated for special purposes e.g. video RAM, BIOS & RAM on PCBs.

Expanded memory

This is specialised memory introduced in 1985 to provide extra memory to DOS programs beyond the limit of standard 1MB (1024 KB) address space.
Expanded memory uses address space in reserved memory called **page frame** divided into 16 KB pages for DOS programs.

Note:

- Expanded memory is configured to use a specific standard called **EMS (Expanded Memory System)**.

- EMS works with DOS to allow access to the extra memory and make it possible to use that memory for certain tasks.
- EMS uses **EMS emulator** (software driver) to hold pages called **page frame** in reserved memory.

Extended memory
-This is a newer alternative to expanded memory, used to provide extra memory to most DOS programs beyond the limit of standard 1MB (1024 KB) address space.
-It is supported by the use of certain types of microprocessors (starting with Intel 80286 and then later ones).
-Also known as **HMA (High Memory Area)** or **HIMEM** by former MS DOS (command line OS).

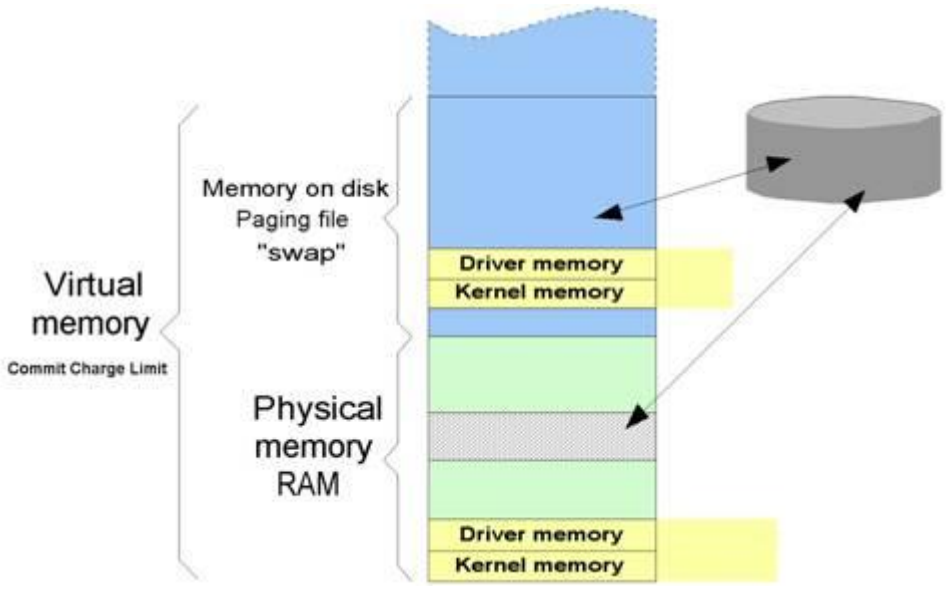
Note:
- Today, most PCs (many programs) use extended memory because newer systems have more powerful microprocessors that render the need to utilize expanded memory obsolete.
- Extended memory does not have to make use of EMS or any other standard, since application programming interface allows communication between the standard and the extended memory.
- In old systems, DOS memory was optimized manually by modifying the **autoexec.bat** and **config.sys** files or using the **MEMMAKER** utility.

Virtual memory
It is a technique in modern OS (e.g. Windows NT) which gives the software the impression that there is more continuous working memory (RAM).
Page file is a section of free hard disk space which is used to continually read and write.
Advantage
Large applications are able run faster and RAM used more efficiently.

Swapping files
-Modern Operating System (e.g. Windows NT) use hard disk space as **virtual memory** extension RAM.
-The space used on hard disk is called **swap file** or **swap space** or **page file**.
-The least recently used files in RAM are swapped out to hard disk until they are needed later and new files are swapped into RAM.

Note:
- The units moved in large OS are called **pages** and swapping is called **paging**.
- Windows and Unix-based OS provide a default swap file of a certain size that the user (system administrator) can usually change.

Advantage
Swap file can be organised as a single continuous space so that fewer I/O operations are required to read or write a complete.



4.5 Troubleshooting memory problems

Troubleshooting involves finding 3 steps things:

1. The problem 2.The cause of the problem 3.The solution to the problem.

Some problems associated with memory failure

1. PC does not boot (load OS).
2. PC boots but reports errors.
3. PC hangs up or reboots.
4. Continuous beeping.
5. Continuous RAM counting during booting.
6. Blue screen with no display.
7. Totally no video display.

Some causes of memory failure & their solutions

1. Too little physical memory (RAM) -Upgrade (add more) RAM.
2. Loose RAM module -Reseat RAM module.
3. Dirty memory slot or RAM pins (conductors) -Clean slots & pins. How?
4. Faulty RAM module -Swap (replace) RAM module.
5. Old version of BIOS -Update BIOS.
6. Computer virus attack -Install & update antivirus regularly.

Troubleshooting guidelines

1. POST
   -Tests and counts memory (RAM) and if different from previous one, issues error message.
   -Error messages depend on type of BIOS and are of 2 types:
   (a) Audio i.e. number of beeps heard (b) Text message shown on the screen.
2. CMOS/BIOS setup
   -Compare memory (RAM) count in POST to installed memory in setup, and if not equal it means memory failure or problem.
3. Other options
   -Include: PC technician software, Norton's utilities and diagnostic disk.

Upgrading memory (RAM)

1. Align RAM module in RAM socket i.e. notch matches alignment key in socket.
2. Hold top edges (not chips on module) and press it into the position in socket.
3. Lock the taps (clips) at the sides to hold it firmly into position in socket.
4. Check CMOS/BIOS setup whether it has been seen (recognised).

Note:

- Since most computers have more than one RAM module, determine which one should be replaced.
- You can also systematically replace one RAM module at a time until the memory error goes away.
- You must acquire the right RAM and correct configuration for the given PC.

See website of PC for capacity of RAM recommended.